Chemistry Notes

Determining Lewis Structures

1. **Step**
   Calculate the total number of valence electrons in the molecule or ion. For ions:
   - Subtract electrons for cations (e.g., K^+)
   - Add electrons for anions (e.g., Cl^-)

2. **Step**
   Identify the central atom, usually the one with the lowest electronegativity. Hydrogen and halogens are typically terminal atoms.
   Note: this rule has exceptions. For instance, in ClO_2, chlorine ist he central atom. If you choose the wrong central atom, it will not be possible to draw a valid structure.

3. **Step**
   Arrange the atoms using single bonds

4. **Step**
   Subtract the electrons used in single bonds from the total valence electron count.

5. **Step**
   Distribute the remaining electrons so that all atoms achieve an octet (except hydrogen, which followst he duet rule). Start assigning lone pairs tot he most electronegative atoms.

6. **Step**
   If there are not enough electrons for all atoms to reach octets, form double or triple bonds.

7. **Step**
   Calculate formal charges to ensure that as many atoms as possible have a formal charge of 0.

Determining Molecular Shape (VSEPR model)

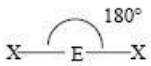
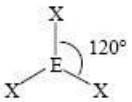
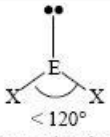
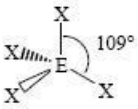
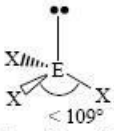
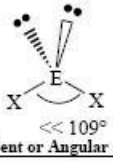
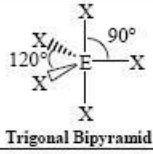
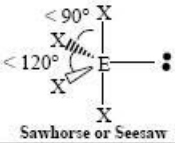
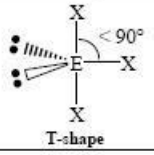
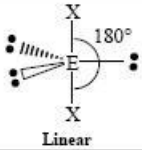
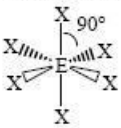
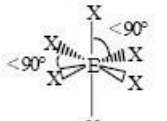
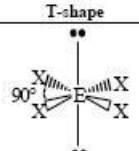
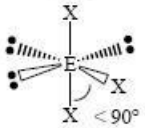
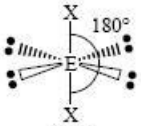
1. Count the number of electron domains (bonding and nonbonding) around the central atom. This number is called the *steric number*.
2. Identify the base geometry depending on the steric number (see VSEPR reference chart)
3. Determine which atoms are replaced by lone electron pairs.
4. Define the final molecular geometry and the approximate bond angles.

   Bond angle reduction rules:

   - ~2° reduction per lone pair for tetrahedral structures
   - ~4° reduction per lone pair for trigonal planar structures

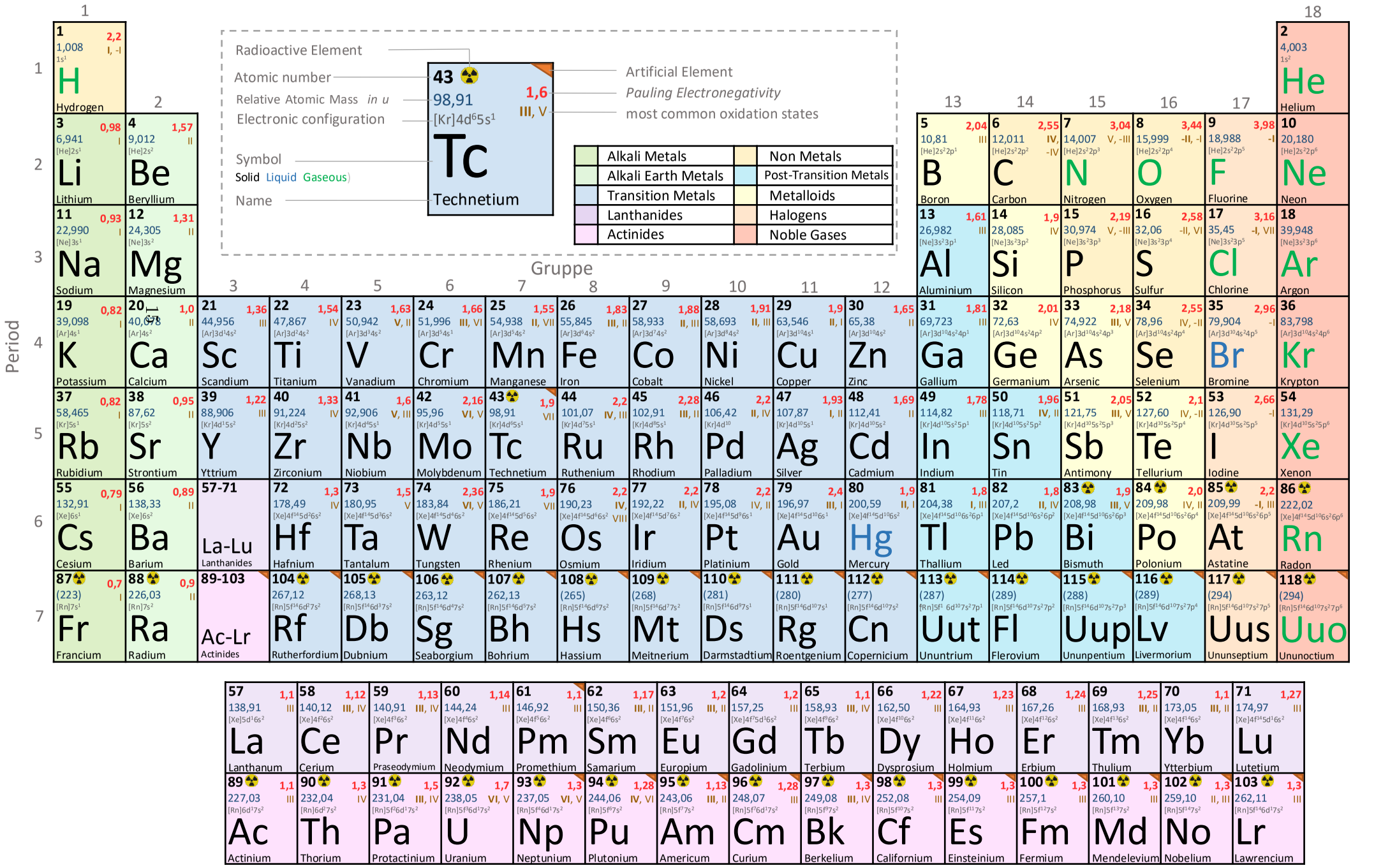
Note: the overall charge of the ion does not influence its VSEPR geometry!

VSEPR Geometries

Steric No.	Basic Geometry 0 lone pair	1 lone pair	2 lone pairs	3 lone pairs	4 lone pairs
2	 Linear				
3	 Trigonal Planar	 Bent or Angular			
4	 Tetrahedral	 Trigonal Pyramid	 Bent or Angular		
5	 Trigonal Bipyramid	 Sawhorse or Seesaw	 T-shape	 Linear	
6	 Octahedral	 Square Pyramid	 Square Planar	 T-shape	 Linear

VSEPR Structures

PERIODIC TABLE OF ELEMENTS



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Determining Oxidation Numbers

Determining oxidation numbers can be complex, but several guiding rules apply:

1. The oxidation number of an atom in ist elemental form is always 0 (e.g. O_2, N_2, H_2, Na(s))
2. The oxidation number of a monoatomic ion equals ist charge (e.g. $Na^+ \rightarrow +1, Cl^- \rightarrow -1$)
3. Hydrogen is +I, except in hydrides (-I)
4. Oxygen is -II, except in peroxides (-I)
5. Alkali metals (group I) are always +I
6. Alkaline earth metals (group II) are always +II
7. The sum of oxidation numbers in a neutral molecule equals 0. In a polyatomic ion, the sum equals the ionic charge.
8. Covalent bonds are treated as if ionic for oxidation-number assignment, electrons are attributed to the more electronegative atom.
9. A positive oxidation number cannot exceed the group number of the element.
10. The maximum negative oxidation number is -VIII (group number 8)
11. Elements in the second main group: +II
12. Elements in the third main group (except Tl): +III

Determining the Remaining Oxidation Numbers

The determination of the remaining oxidation numbers is carried out by balancing the individual molecules and adjusting the oxidation states of the other atoms accordingly. The total charge of a molecule must always correspond to its overall ionic charge. Therefore, neutral molecules must have a sum of oxidation numbers equal to zero.

Common Oxidation Numbers

Atom	Oxidation Numbers
H	**I**,-I
O	**-II**,-I
C	**-IV,+IV**,+II
Zn	**+II**
Pb	**+II**,+IV
S	**+VI**,+IV,+II,-II
N	**-III**,+V,+IV,+III,+II
P	**+V**,+III,-III
F	**-I**
Cl	**-I**,+I,+III,+V,+VII
Br	**-I**,+I,+III,+V,+VII
I	**-I**,+I,+III,+V,+VII
At	**-I**,+I,+III,+V,+VII
Kr	**+II**
Xe	**+II**,+IV,+VI

Steps to Balance Redox Equations

1. Write the incomplete (skeleton) equation with reactants and products.
2. Determine the oxidation numbers of all atoms. Identify which are oxidized and which are reduced.
3. Split the process into half-reactions: one for oxidation and one for reduction.
4. If needed, balance the stoichiometric coefficients of atoms involved in redox changes.
5. Write down the number of electrons lost or gained in each half-reaction.
6. Balance charges using H_3O^+ (acidic or neutral solutions) or OH^- (basic solutions).
7. Balance water (H_2O) molecules to complete atom balance.
8. Equalize electron numbers by multiplying each half-reaction by appropriate factors.
9. Add the half-reactions together to obtain the full balanced equation.

Table of Acids with Ka and pKa Values*

CLAS

Acid	HA	A^-	Ka	pKa	Acid Strength	Conjugate Base Strength
Hydroiodic	HI	I^-	Strong acids completely dissociate in aq solution (Ka > 1, pKa < 1). Conjugate bases of strong acids are ineffective bases.			
Hydrobromic	HBr	Br^-				
Perchloric	$HClO_4$	ClO_4^-				
Hydrochloric	HCl	Cl^-				
Chloric	$HClO_3$	ClO_3^-				
Sulfuric (1)	H_2SO_4	HSO_4^-				
Nitric	HNO_3	NO_3^-				
Hydronium ion	H_3O^+	H_2O	1	0.0	↑	↓
Iodic	HIO_3	IO_3^-	1.6×10^{-1}	0.80		
Oxalic (1)	$H_2C_2O_4$	$HC_2O_4^-$	5.9×10^{-2}	1.23		
Sulfurous (1)	H_2SO_3	HSO_3^-	1.54×10^{-2}	1.81		
Sulfuric (2)	HSO_4^-	SO_4^{2-}	1.2×10^{-2}	1.92		
Chlorous	$HClO_2$	ClO_2^-	1.1×10^{-2}	1.96		
Phosphoric (1)	H_3PO_4	$H_2PO_4^-$	7.52×10^{-3}	2.12		
Arsenic (1)	H_3AsO_4	$H_2AsO_4^-$	5.0×10^{-3}	2.30		
Chloroacetic	$CH_2ClCOOH$	CH_2ClCOO^-	1.4×10^{-3}	2.85		
Citric (1)	$H_3C_6H_5O_7$	$H_2C_6H_5O_7^-$	8.4×10^{-4}	3.08		
Hydrofluoric	HF	F^-	7.2×10^{-4}	3.14		
Nitrous	HNO_2	NO_2^-	4.0×10^{-4}	3.39		
Formic	HCOOH	$HCOO^-$	1.77×10^{-4}	3.75		
Lactic	$HCH_3H_5O_3$	$CH_3H_5O_3^-$	1.38×10^{-4}	3.86		
Ascorbic (1)	$H_2C_6H_6O_6$	$HC_6H_6O_6^-$	7.9×10^{-5}	4.10		
Benzoic	C_6H_5COOH	$C_6H_5COO^-$	6.46×10^{-5}	4.19		
Oxalic (2)	$HC_2O_4^-$	$C_2O_4^{2-}$	6.4×10^{-5}	4.19		
Hydrazoic	HN_3	N_3^-	1.9×10^{-5}	4.72		
Citric (2)	$H_2C_6H_5O_7^-$	$HC_6H_5O_7^{2-}$	1.8×10^{-5}	4.74		
Acetic	CH_3COOH	CH_3COO^-	1.76×10^{-5}	4.75		
Propionic	CH_3CH_2COOH	$CH_3CH_2COO^-$	1.34×10^{-5}	4.87		
Pyridinium ion	$C_5H_4NH^+$	C_5H_4N	5.6×10^{-6}	5.25		
Citric (3)	$HC_6H_5O_7^{2-}$	$C_6H_5O_7^{3-}$	4.0×10^{-6}	5.40		
Carbonic (1)	H_2CO_3	HCO_3^-	4.3×10^{-7}	6.37		
Sulfurous (2)	HSO_4^-	SO_4^{2-}	1.02×10^{-7}	6.91		
Arsenic (2)	$H_2AsO_4^-$	$HAsO_4^{2-}$	$8/9.3 \times 10^{-8}$	7.10/7.03		
Hydrosulfuric	H_2S	HS^-	$1.0 \times 10^{-7}/9.1 \times 10^{-8}$	7/7.04		
Phosphoric (2)	$H_2PO_4^-$	HPO_4^{2-}	6.23×10^{-8}	7.21		
Hypochlorous	HClO	ClO^-	$3.5/3.0 \times 10^{-8}$	7.46/7.53		
Hypobromous	HBrO	BrO^-	2×10^{-9}	8.70		
Hydrocyanic	HCN	CN^-	6.17×10^{-10}	9.21		
Boric (1)	H_3BO_3	$H_2BO_3^-$	5.8×10^{-10}	9.23		
Ammonium ion	NH_4^+	NH_3	5.6×10^{-10}	9.25		
Phenol	C_6H_5OH	$C_6H_5O^-$	1.6×10^{-10}	9.80		
Carbonic (2)	HCO_3^-	CO_3^{2-}	4.8×10^{-11}	10.32		
Hypoiodous	HIO	IO^-	2×10^{-11}	10.70		
Arsenic (3)	$HAsO_4^{2-}$	AsO_4^{3-}	$6.0 \times 10^{-10}/3.0 \times 10^{-12}$	9.22/11.53		
Hydrogen peroxide	H_2O_2	HO_2^-	2.4×10^{-12}	11.62		
Ascorbic (2)	$HC_6H_6O_6^-$	$C_6H_6O_6^{2-}$	1.6×10^{-12}	11.80		
Phosphoric (3)	HPO_4^{2-}	PO_4^{3-}	$4.8/2.2 \times 10^{-13}$	12.32/12.66		
Water	H_2O	OH^-	1.0×10^{-14}	14.0		
Group I metal hydroxides (LiOH, NaOH, etc.)			Strong bases completely dissociate in aq solution (Kb > 1, pKb < 1). Conjugate acids (cations) of strong bases are ineffective bases.			
Group II metal hydroxides ($Mg(OH)_2$, $Ba(OH)_2$, etc.)						

* Compiled from Appendix 5 Chem 1A, B, C Lab Manual and Zumdahl 6th Ed. The pKa values for organic acids can be found in Appendix II of Bruice 5th Ed.

Table O

Standard Electrochemical Potentials of Selected Substances

T = 298,15 K p = 1,00 bar a(Ionen) = 1 mol/kg

Tabelle O1 Metals

Reduced Form	Oxidized Form	Electron transfer	E_0 / Volt
K	K^+	1	-2,92
Ba	Ba^{2+}	2	-2,9
Sr	Sr^{2+}	2	-2,89
Ca	Ca^{2+}	2	-2,87
Na	Na^+	1	-2,71
Mg	Mg^{2+}	2	-2,36
Al	Al^{3+}	3	-1,66
Mn	Mn^{2+}	2	-1,18
Cr	Cr^{2+}	2	-0,91
Zn	Zn^{2+}	2	-0,76
Cr	Cr^{3+}	3	-0,74
Fe	Fe^{2+}	2	-0,41
Co	Co^{2+}	2	-0,28
Ni	Ni^{2+}	2	-0,23
Sn	Sn^{2+}	2	-0,14
Pb	Pb^{2+}	2	-0,13
Fe	Fe^{3+}	3	-0,02
½ H_2	H^+	1	0
Cu	Cu^{2+}	2	0,35
Cu	Cu^+	1	0,52
Ag	Ag^+	1	0,8
Hg	Hg^{2+}	2	0,85
Pt	Pt^{2+}	2	1,2
Au	Au^+	1	1,68

Table O2 Non metals

Reduced Form	Oxidized Form	Electron transfer	E_0 / Volt
Te^{2-}	Te (s)	2	-0,95
Se^{2-}	Se (s)	2	-0,92
S^{2-}	S (s)	2	-0,48
J^-	½ J_2 (s)	1	0,54
Br^-	½ Br_2 (l)	1	1,07
Cl^-	½ Cl_2 (g)	1	1,36
F^-	½ F_2 (g)	1	2,87

Standard Electrochemical Potentials of Selected Substances

T = 298,15 K p = 1,00 bar a(ions) = 1 mol/kg

Table O3 Ion exchanges

Reduced Form	Oxidized Form	Electron transfer	E_0 / Volt
Cr^{2+}	Cr^{3+}	1	-0,41
Sn^{2+}	Sn^{4+}	2	0,15
Cu^{+}	Cu^{2+}	1	0,17
$Fe(CN)_6^{4-}$	$Fe(CN)_6^{3-}$	1	0,36
MnO_4^{2-}	MnO_4^{-}	1	0,58
Fe^{2+}	Fe^{3+}	1	0,77
Pb^{2+}	Pb^{4+}	2	1,66

Table O4 Redox Reactions

Reduced Form	Oxidized Form	Electron transfer	E_0 / Volt
$H_2 + 2\ OH^-$	$2\ H_2O$	2	-0,83
$Pb + SO_4^{2-}$	$PbSO_4$	2	-0,36
$4\ OH^-$	$2\ H_2O + O_2$	4	0,4
$NO_2 + H_2O$	$NO_3^- + 2\ H^+$	1	0,8
$½\ O_2 + 2\ H^+$	H_2O	2	0,83
$NO + 2\ H_2O$	$NO_3^- + 4\ H^+$	3	0,96
$Mn^{2+} + 2\ H_2O$	$MnO_2 + 4\ H^+$	2	1,23
$Pb^{2+} + 2\ H_2O$	$PbO_2 + 4\ H^+$	2	1,46
$Mn^{2+} + 4\ H_2O$	$MnO_4^- + 8\ H^+$	5	1,51
$PbSO_4 + 2\ H_2O$	$PbO_2 + 4\ H^+ + SO_4^{2-}$	2	1,69
$MnO_2 + 2\ H_2O$	$MnO_4^- + 4\ H^+$	3	1,7
$2\ H_2O$	$H_2O_2 + 2\ H^+$	2	1,77
$O_2 + H_2O$	$O_3 + 2\ H^+$	2	2,07